

SIGNAL

User Operations Manual

AI-Powered Energy Intelligence Platform

Shell Energy · Enterprise Crude Oil Market Intelligence

Version 2.0 · Analyst & Executive Edition

Classification: INTERNAL / CONFIDENTIAL

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1.1 What SIGNAL Does

SIGNAL (System for Intelligent Geopolitical News and Asset Linking) is Shell Energy's enterprise crude oil market intelligence platform. It ingests real-time geopolitical news, applies NLP sentiment analysis, and runs two machine-learning models to deliver actionable price-direction and price-level signals to traders, analysts, and executives.

SIGNAL is a **decision-support system** — not an autonomous trading engine. It augments human judgment with data-driven intelligence, enabling faster and more informed responses to market-moving events.

1.2 Dual-Engine Architecture

Engine	Type	Output	Performance
Direction Signal Engine	Classification (Random Forest + XGBoost)	UP ▲ / DOWN ▼ + confidence probability	81.5% accuracy (holdout test set)
Price Forecast Engine	Regression (Stacking: Ridge+HistGBM +LightGBM → RidgeCV)	Next-day WTI price in \$/bbl	RMSE \$2.07/bbl R ² 0.953 MAPE ~2.5%

Key Principle:

Both engines run daily after market close. Results publish to the dashboard the following morning, giving users overnight intelligence before trading opens.

2.1 Login Process

Navigate to the SIGNAL URL. Enter your registered email and password, then click **Access SIGNAL**.

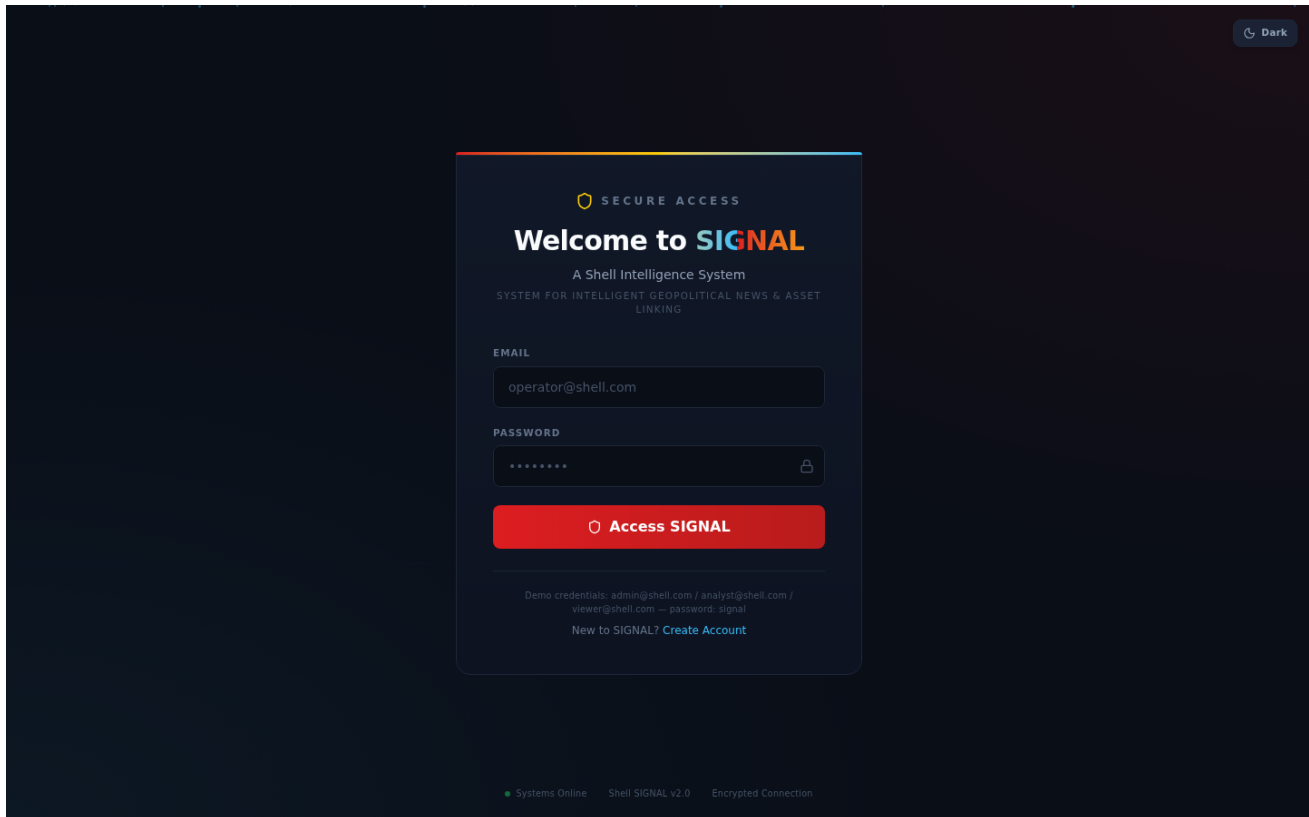


Figure 2-1 — SIGNAL Login Screen

2.2 Registration

New users click **Create Account** on the login page. Enter full name, email, a secure password, and your role access code, then click **Create Account**.

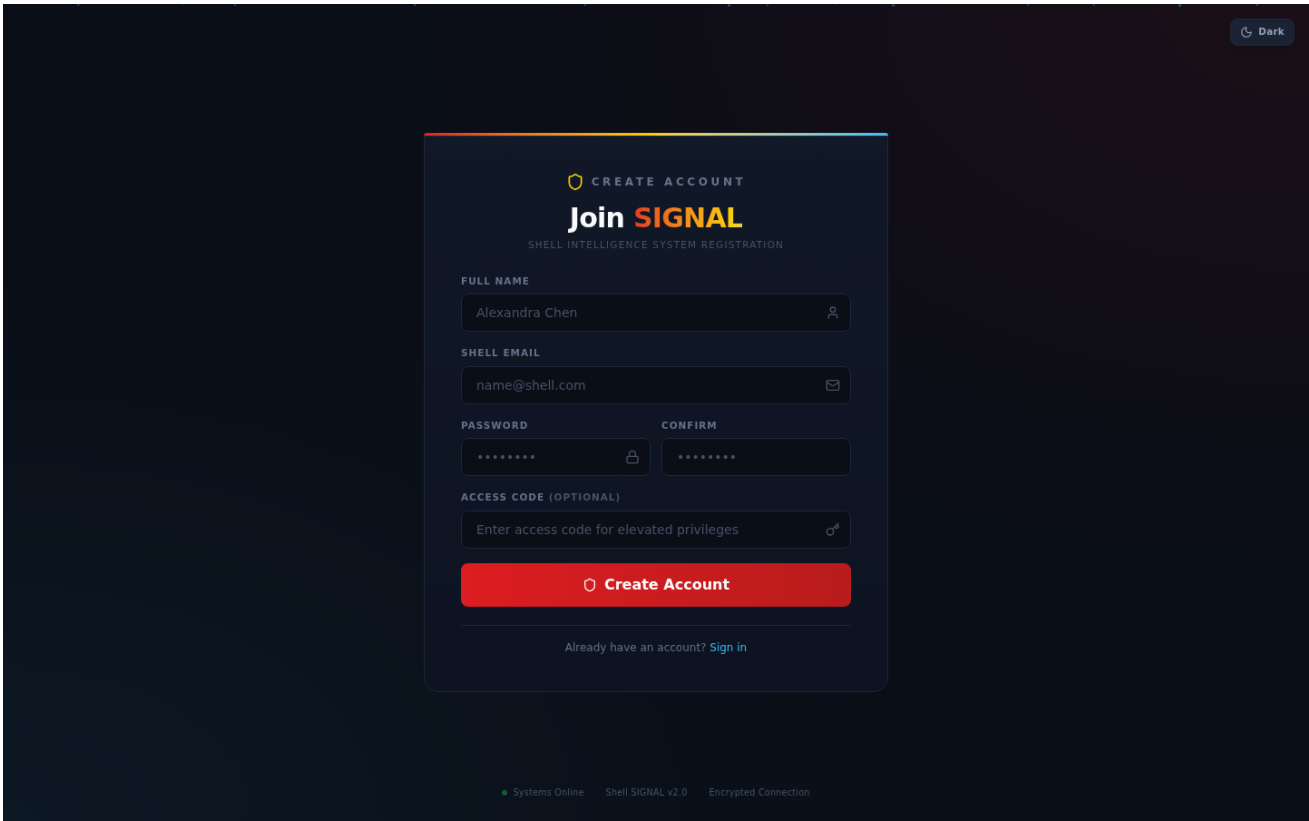


Figure 2-2 — SIGNAL Registration Screen

2.3 Role-Based Access

Role	Access Code	Pages	Special Features
Analyst	SIGNAL-ANL-01	Command Center, Intelligence Feed, Commodities, Analytics, Directives	Standard Analyst Role, Price Forecast
Executive	SIGNAL-EXEC-01	All Analyst pages + executive onboarding	Gold EXEC badge; Strategic briefing; Escalation pr
Viewer	(no code)	Command Center + Intelligence Feed (read-only)	No model/config access

2.4 Onboarding Flow

Step	Analyst (5 Steps)	Executive (6 Steps)
1	Welcome & platform overview	Cinematic security verification
2	Alert threshold configuration	Strategic briefing with live KPI data
3	Data source acknowledgement	Strategic focus area selection
4	Dashboard preview	Escalation protocols (cadence + level)
5	Activation — access granted	Dashboard preview
6	—	Activation — EXEC badge applied

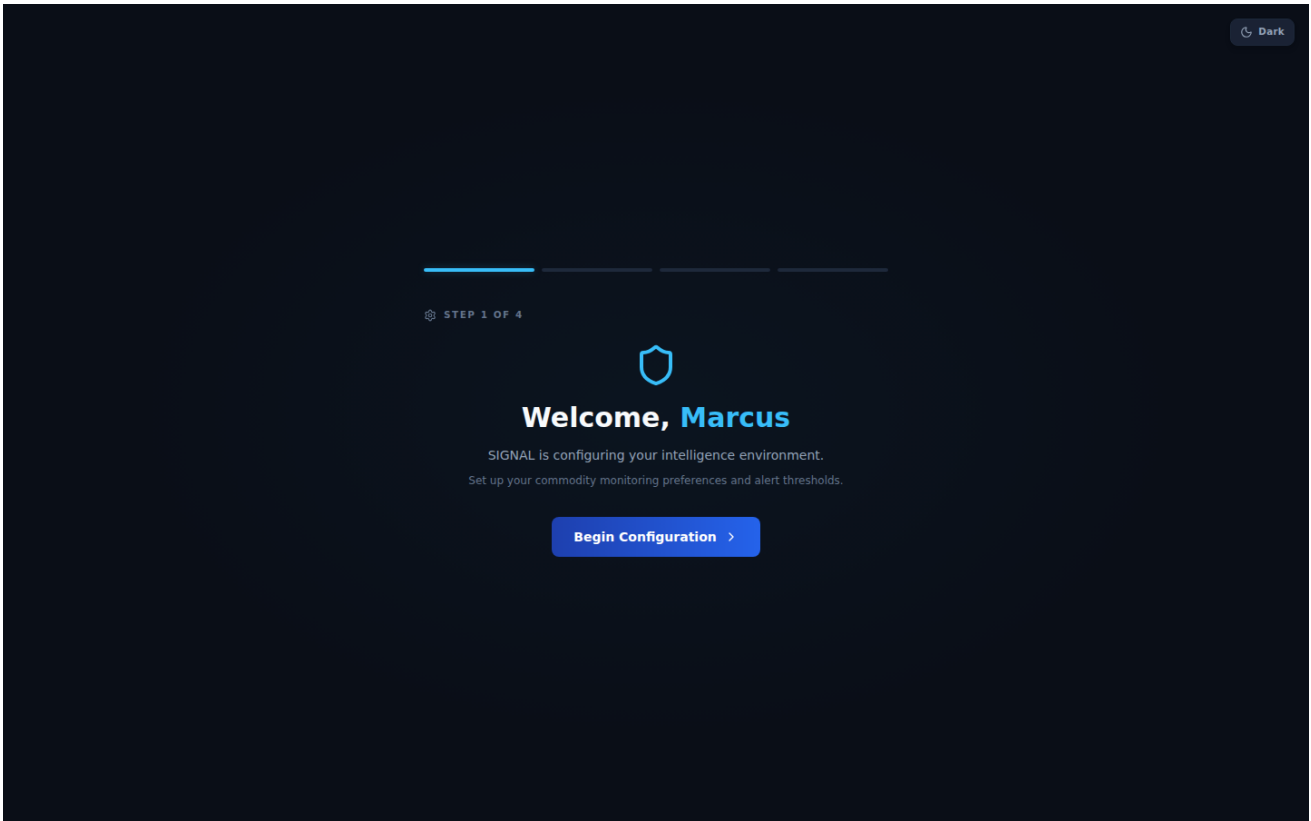


Figure 2-3 — Onboarding Flow

3 Dashboard Walkthrough

The Command Center (/dashboard) is the primary intelligence screen. It delivers a complete situational picture — current AI signal, live price, model performance, news flow, and alerts — on a single view.



Figure 3-0 — Full Command Center Overview

3.1 KPI Summary Cards

Four metric cards at the top provide the critical numbers at a glance:

Card	What It Shows	How to Interpret
WTI Price	Latest West Texas Intermediate spot price (\$/Bbl)	Benchmark for all model outputs and alerts.
Today's Signal	AI direction: UP / DOWN + confidence %	Confidence > 75% = strong actionable signal.
24h Change	Absolute (\$) and % change vs. prior close	Quick gauge of intraday momentum.
Sentiment Score	Aggregated 24-hour news sentiment (–1.0 to +1.0)	> 0.3 bullish · < –0.3 bearish · otherwise neutral.

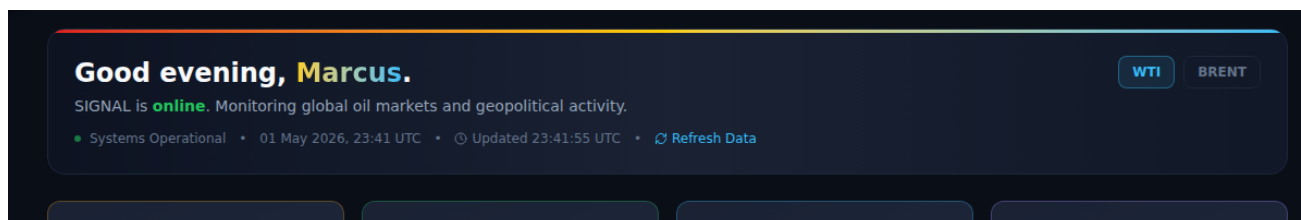


Figure 3-1 — KPI Summary Cards

3.2 Holdout Validation Chart

The left panel shows the price forecast model's performance on unseen test data. Three lines plotted: **Actual** (real WTI close), **Predicted** (model forecast), **Baseline** (naïve last-price benchmark). When Predicted tracks closer to Actual than Baseline, the model adds value.

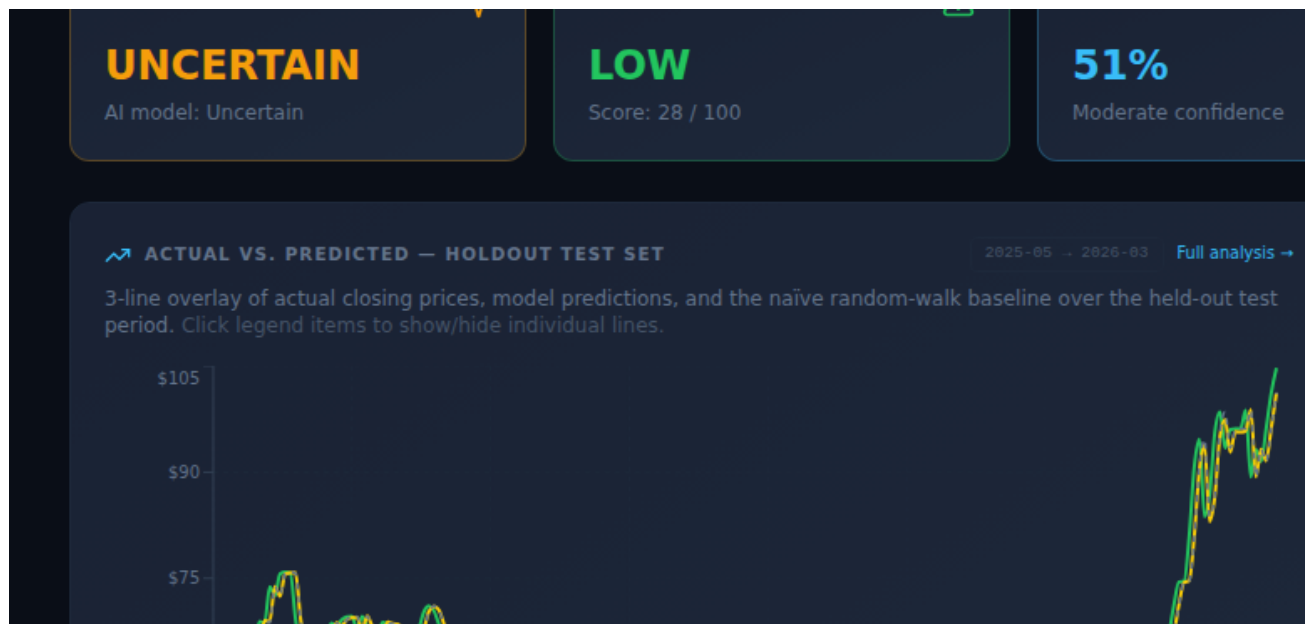


Figure 3-2 — Holdout Validation Chart

3.3 Intelligence Feed Panel

The right-side panel shows the five most recent crude-market headlines with sentiment badges (Positive / Negative / Neutral), source, timestamp, and link to full article.

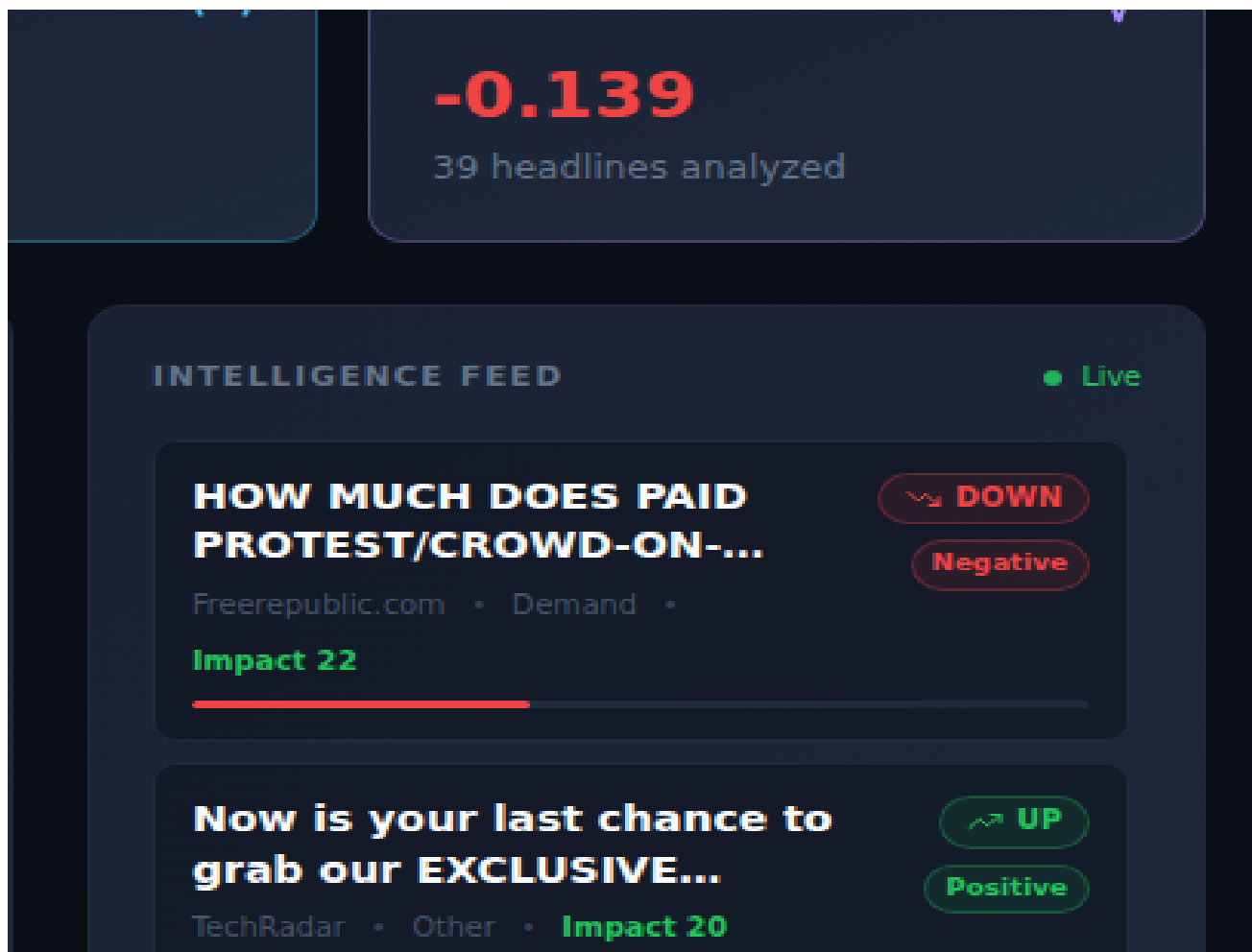


Figure 3-3 — News Intelligence Panel

3.4 Live WTI Price Chart & Active Alerts

Lower-left: full WTI price history with date-range controls (1W/1M/3M/1Y/All). Lower-right: recently triggered alerts, severity-coded Critical (red), Warning (amber), or Info (blue).

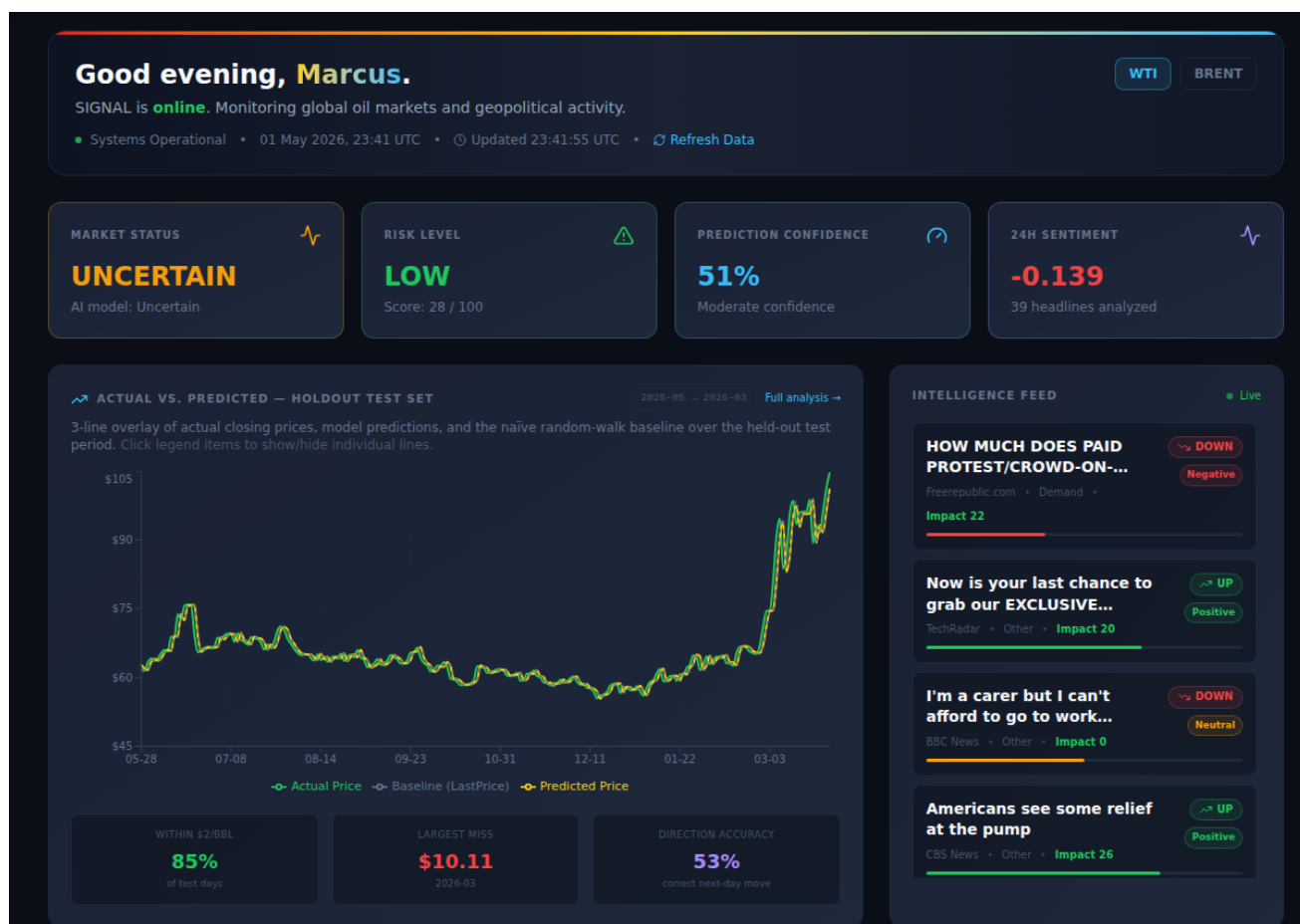


Figure 3-4 — Price Chart + Alerts Panel

4 Intelligence Feed (/signals)

The Intelligence Feed is the primary research view for analysts. It surfaces all ingested headlines with full NLP enrichment.

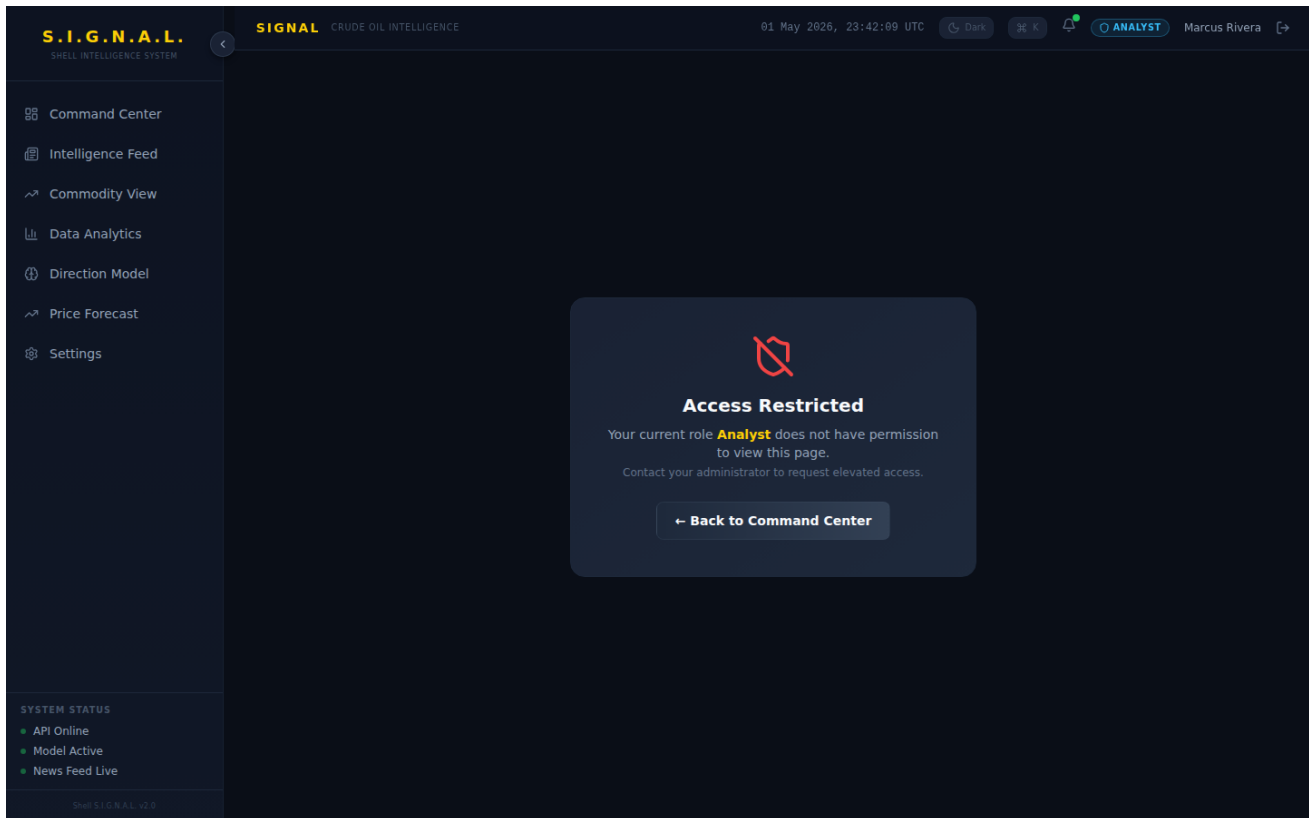


Figure 4-0 — Intelligence Feed Overview

4.1 Headline Table

Column	Description
Date/Time	When the article was published (local timezone)
Headline	Article title — click to open the source URL
Source	Publication name (e.g., Reuters, Bloomberg)
Sentiment	NLP-derived: Positive / Negative / Neutral badge
Score	Raw VADER sentiment score (–1.0 to +1.0)
AI Impact	Model's assessment of directional impact on WTI

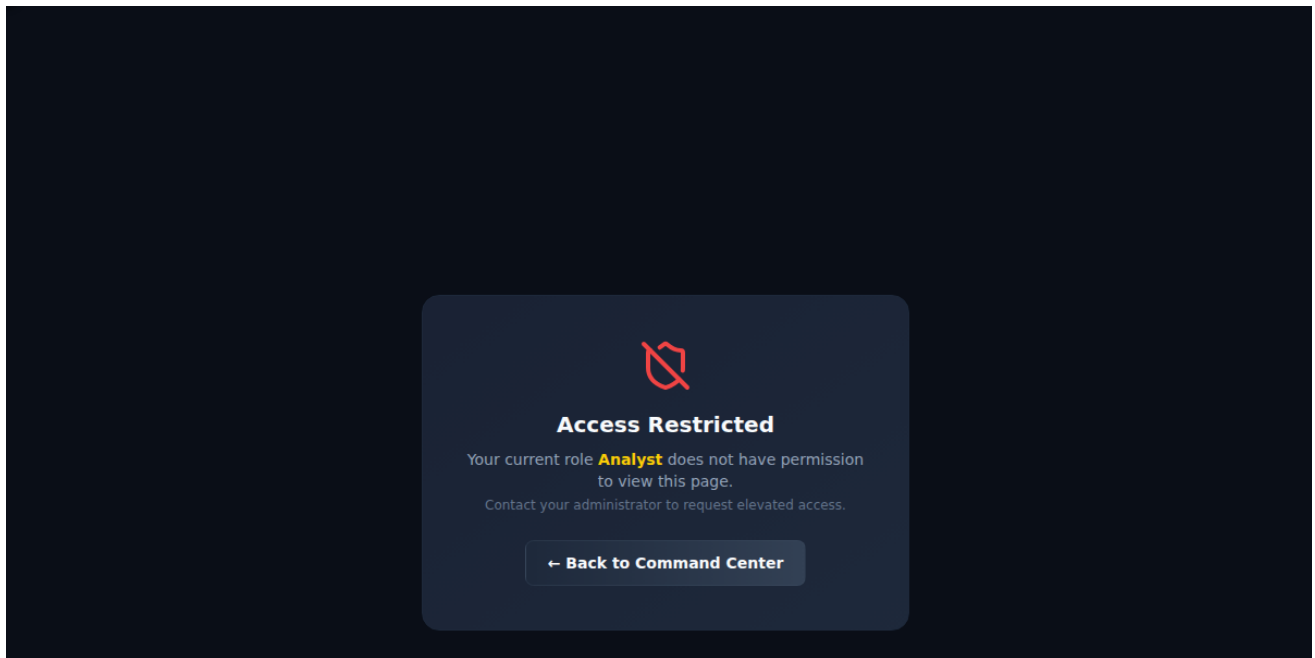


Figure 4-1 — Headline Table

4.2 Sentiment Scoring

SIGNAL uses VADER (Valence Aware Dictionary and sEntiment Reasoner). Each headline receives a compound score: Positive ($> +0.05$), Negative (< -0.05), Neutral.

4.3 Search & Filters

The search bar performs full-text search across all stored headlines. Filter chips narrow results by sentiment label or date range.

5

Commodity View (/commodity)

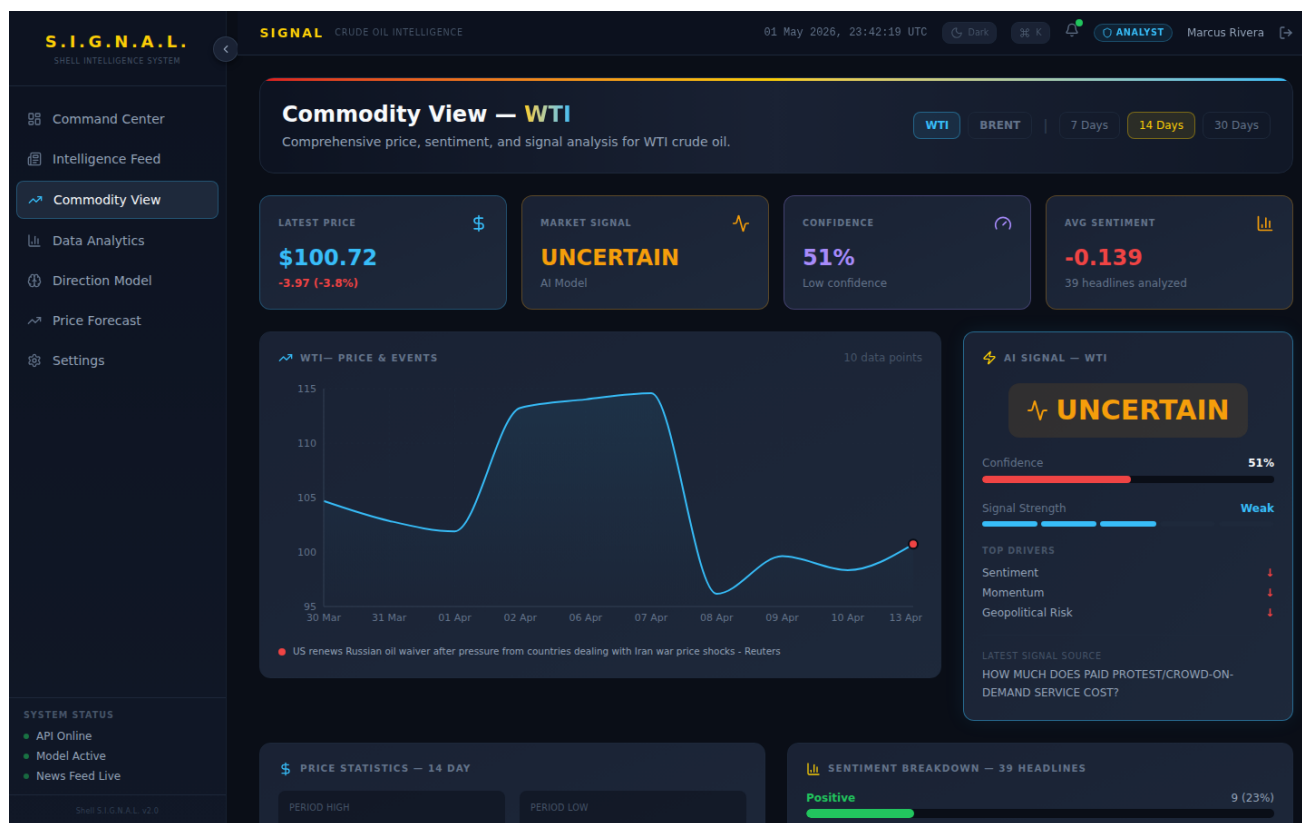


Figure 5-0 — Commodity View Overview

5.1 Price Chart

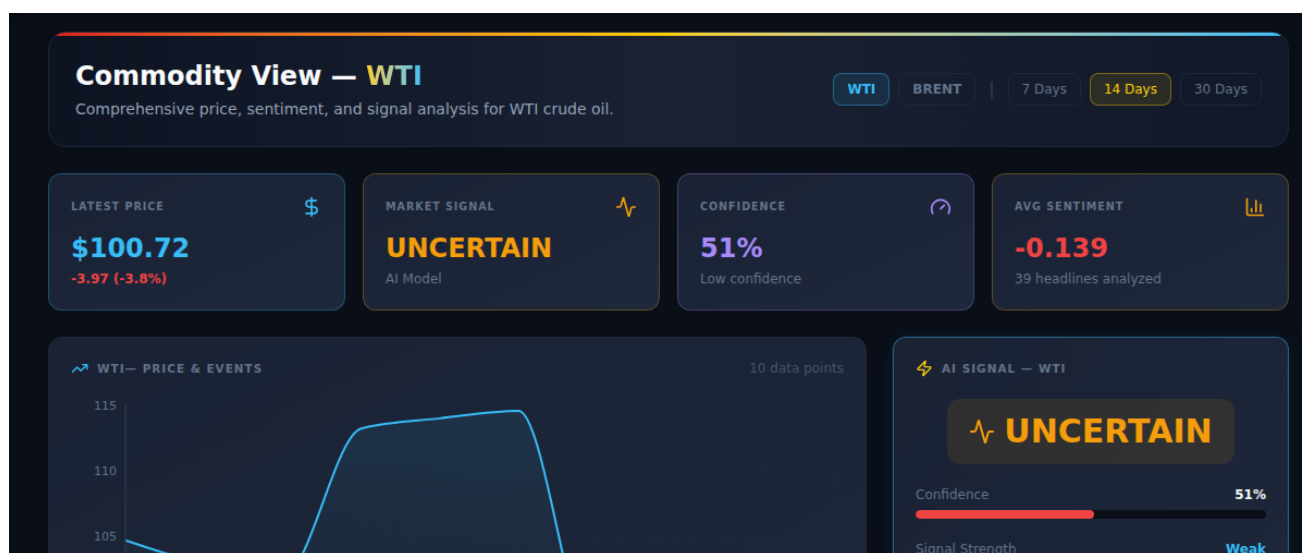


Figure 5-1 — WTI Price Chart with MA Overlays

5.2 Moving Average Overlays

- 7-day MA — Short-term trend

- **30-day MA** — Medium-term momentum
- **90-day MA** — Long-term cycle context

5.3 Summary Statistics

Statistic	Description
52-Week High / Low	Extreme prices over the last 52 weeks
YTD Performance	Percentage return since January 1
Avg Daily Range	Mean intraday high-minus-low over the last 30 days
30-Day Volatility	Annualised standard deviation of daily returns

6 Analytics (/analytics)

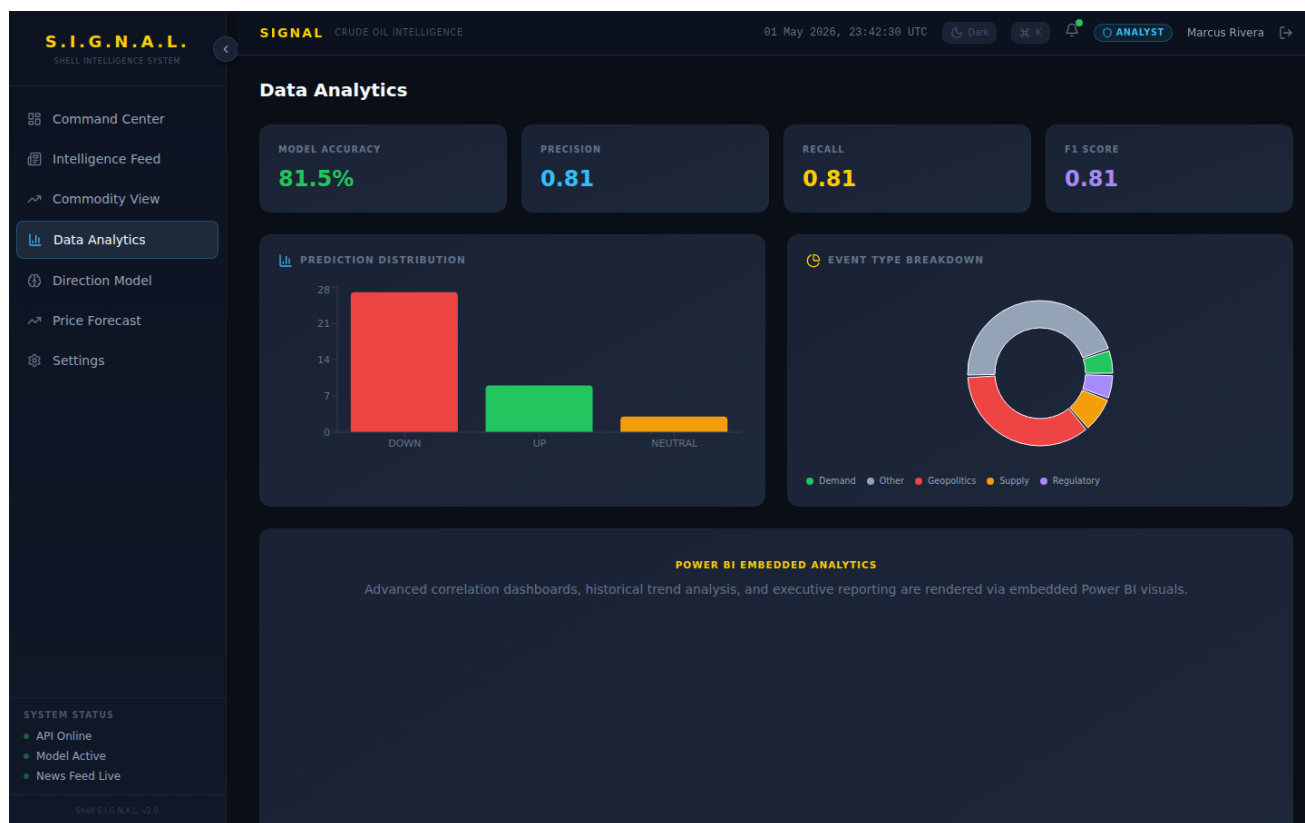


Figure 6-0 — Analytics Page Overview

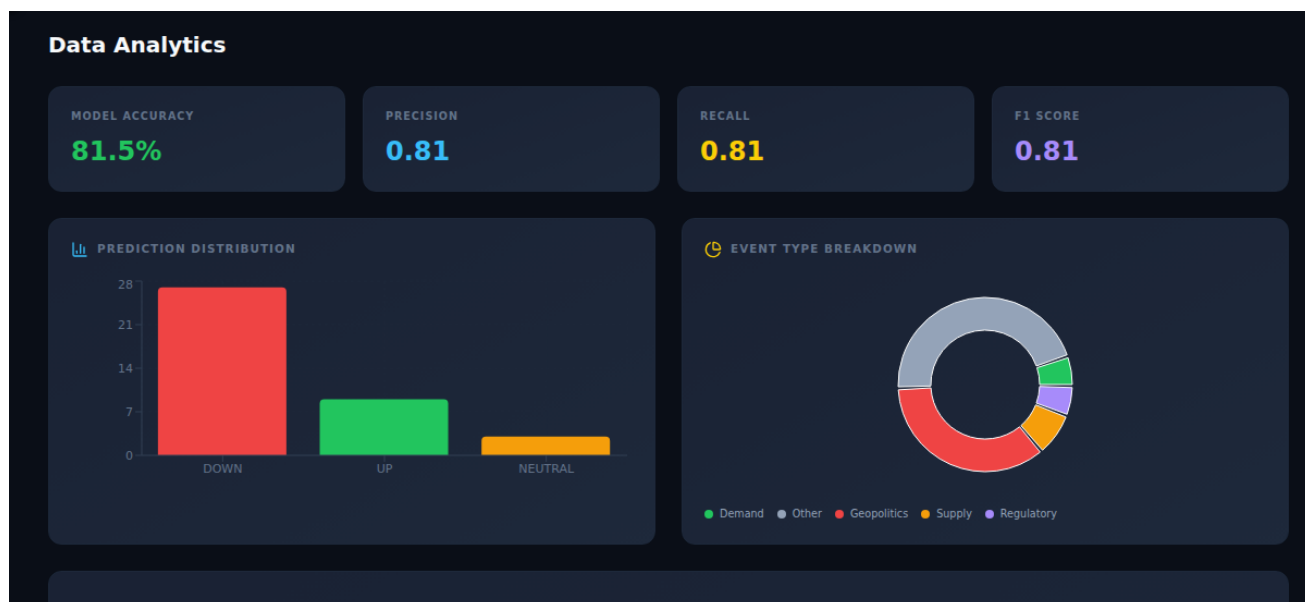


Figure 6-1 — Analytics Charts Detail

6.1 Sentiment Trend

Daily average sentiment over 30/90/365-day windows. Sustained negative sentiment often precedes price weakness; sharp reversals are potential inflection signals.

6.2 Article Volume

Headlines ingested per day. Volume spikes indicate high-information days (OPEC decisions, supply disruptions) where signal confidence is typically higher.

6.3 Signal Confidence Over Time

Time-series of the direction model output probability. Extended low-confidence periods may indicate regime change or an unusually noisy news environment.

7 AI Models Explained

7.1 Direction Signal Engine

Component	Detail
Algorithm	Random Forest + XGBoost ensemble
Input features	10-day price lags, 3/5/10/20-day MAs, 5/10/20-day volatility, rolling sentiment & article-volume averages, d
Output	UP or DOWN label + probability (0–1)
Gating	Displayed as 'Uncertain' when probability < 0.60
Accuracy	81.5% on holdout test set



Figure 7-1 — Direction Model Dashboard

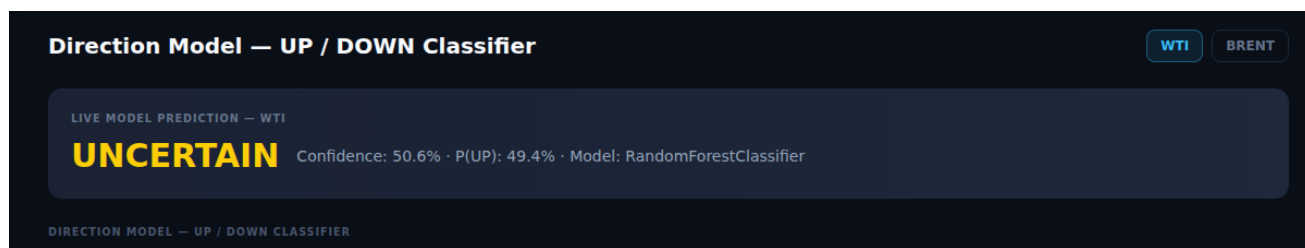


Figure 7-2 — Direction Model Metric Cards

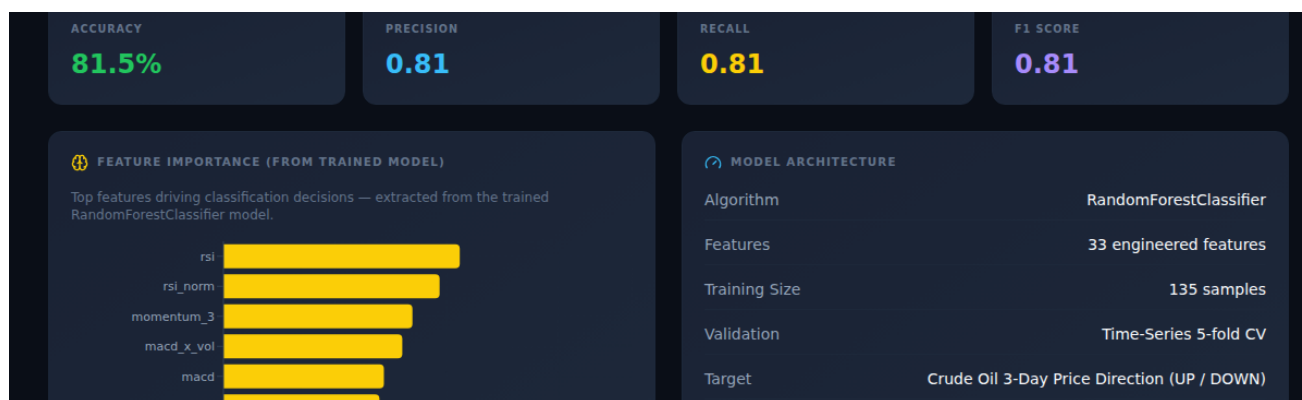


Figure 7-3 — Prediction History Chart

7.2 Price Forecast Engine

Component	Detail
Architecture	Stacking: Ridge + HistGradientBoosting + LightGBM (base) → RidgeCV (meta)
Input features	Same enriched feature set (1,270 daily rows)
Output	Predicted next-day WTI price in \$/bbl
RMSE	\$2.07/bbl (vs. naïve baseline \$2.09/bbl — model outperforms)
R ²	0.9532 — explains 95.3% of daily price variance
MAPE	~2.5% average daily error

7.3 Performance in Business Terms

Direction Engine:

If SIGNAL says UP with 82% confidence, that signal has historically been correct ~82% of the time — materially better than a 50/50 coin flip.

Price Forecast:

A \$2.07 RMSE at ~\$75/bbl spot = ~2.8% average error. The model's daily price estimate is typically within \$2 of the actual close.

Important Caveat:

Black swan events fall outside the model's training distribution. Human judgment must override AI outputs during extraordinary geopolitical events.

8 Price Forecast (/forecast)

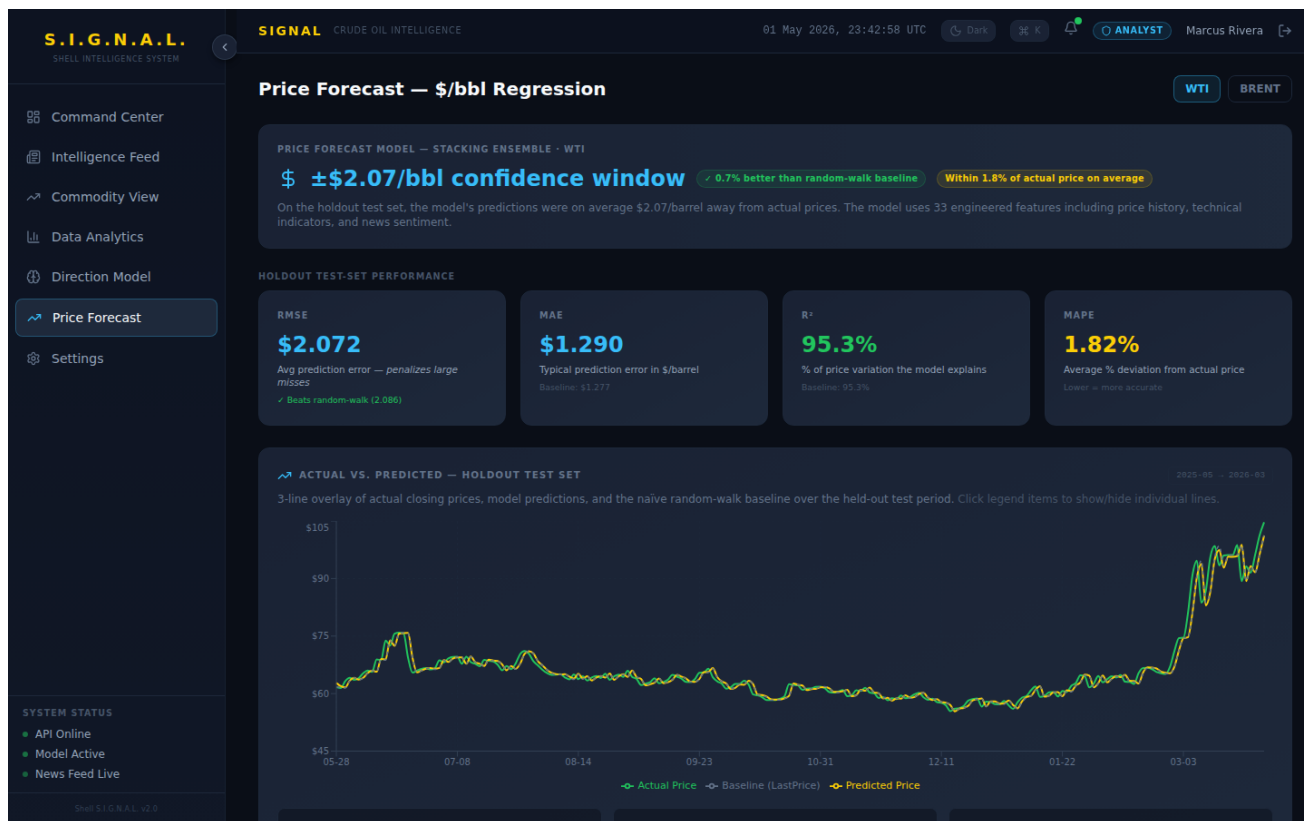


Figure 8-0 — Price Forecast Page Overview

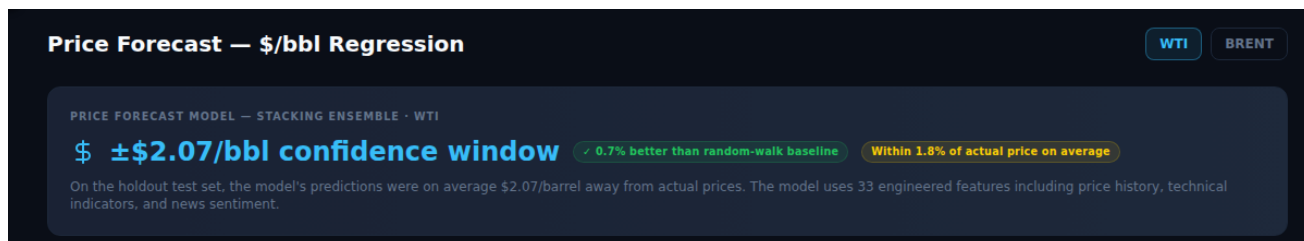


Figure 8-1 — Forecast Metric Cards

8.1 Actual vs. Predicted Chart



Figure 8-2 — Actual vs. Predicted Price Chart

8.2 Error Analysis Row

Card	What It Measures
Within $\pm 2\%$ of Actual	% of holdout days where prediction was within 2% of real price
Largest Single-Day Miss	Maximum absolute error on any single holdout day
Direction Accuracy	% of holdout days where model predicted the correct direction

8.3 Feature Importance

Horizontal bar chart showing which input features drive the model most, colour-coded: price-lag (blue, dominant), moving-average (green), volatility (amber), sentiment (purple).

8.4 Model Architecture

A 3-step explainer shows the stacking pipeline: Base Models $\rightarrow$ Meta-Learner $\rightarrow$ Output. A collapsible model comparison table lists all evaluated algorithms with RMSE and R^2 .

9 Interpreting Signals

9.1 Signal States

Signal State	Confidence	Meaning	Suggested Action
■ Strong UP	> 80%	High-probability bullish move	Lean long; review call optionality
↑ UP	60–80%	Moderate bullish lean	Monitor; wait for news confirmation
Uncertain	< 60%	Model lacks conviction	Hold; avoid directional bets
↓ DOWN	60–80%	Moderate bearish lean	Review short-side hedges
■ Strong DOWN	> 80%	High-probability bearish move	Review put protection or short futures

9.2 Confidence Thresholds

Probability	Label	Interpretation
$p > 0.80$	Strong Signal	High conviction — historically most reliable
0.65–0.80	Moderate Signal	Directional lean — use with corroborating indicators
0.60–0.65	Weak Signal	Low conviction — treat with caution
$p < 0.60$	Uncertain	Model abstains — no actionable signal

9.3 Decision Framework

Scenario	Recommended Response
Strong UP + Positive Sentiment (>0.3) + Rising volatility	High-confidence bullish setup. Strong case for long exposure.
Strong DOWN + Negative Sentiment (<-0.3) + Bearish headlines	High-confidence bearish setup. Review downside hedges.
Uncertain signal, any sentiment	Do not act on signal alone. Wait for next cycle.
Signal contradicts news flow	Signal may be lagged. Apply extra scrutiny; effect may already be priced.
Confidence $> 80\%$ but forecast change $< \$0.50$	High confidence, small expected move. May not justify transaction costs.

10 Use Cases

10.1 Trader / Analyst Workflow

08:00 — Morning check: Open Command Center. Note Today's Signal, confidence, and overnight sentiment.

08:15 — News review: Review Intelligence Feed for high-impact overnight headlines. Cross-reference with the signal direction.

08:30 — Price context: Check Commodity View for overnight price action and MA positioning.

09:00 — Signal decision: If signal > 75% confidence and news confirms direction, escalate to trading desk.

Post-close validation: Compare Price Forecast prediction to actual close. Log divergences for model feedback.

10.2 Risk Management

- Set price alert thresholds in Settings → Alert Configuration.
- Enable sentiment spike notifications for early warning of market stress.
- Use the 30-day volatility metric on Commodity View to calibrate VaR calculations.
- If sustained negative sentiment coincides with a DOWN signal, flag for senior review.

10.3 Procurement & Hedging

Scenario	SIGNAL Insight	Action
Strong DOWN + forecast \$2 drop	Model projects decline with high confidence	Delay spot purchase; wait for lower price
Strong UP + forecast \$3 rise	Model projects increase with high confidence	Accelerate procurement; lock in swap or call option
Uncertain / low confidence	Model has insufficient information	Maintain standard procurement schedule

11 Best Practices

11.1 Using SIGNAL Effectively

✓ Check daily, not hourly	SIGNAL updates once per day after market close. Intraday re-checking adds no new intelligence.
✓ Combine signal + sentiment + news	A strong directional signal with confirming sentiment and clear news catalysts is the highest-quality setup.
✓ Set meaningful alerts	Configure alerts at actual trading thresholds. Over-alerting causes fatigue and desensitises users to genuine signals.
✓ Use the forecast as context	The \$2.07 RMSE means the actual price may be \$2 above or below. Use it to frame the range of expected outcomes.
✓ Review historical performance	Check the Price Forecast page monthly. If RMSE deteriorates materially, re-training may be needed.

11.2 When NOT to Trust Low-Confidence Signals

■ Signal probability < 60%	The model lacks conviction. Treat as 'no signal.'
■ Signal contradicts breaking news	If a major OPEC cut is announced and the signal says DOWN, the news has not yet been incorporated. Wait for the next daily cycle.
■ Consecutive signal reversals	If the signal flips UP/DOWN on consecutive days, the market is in a noisy regime. Widen decision thresholds temporarily.
■ Extraordinary geopolitical events	Black swans fall outside the model's training distribution. Human judgment must override AI outputs in these scenarios.

12 Settings & Configuration (/settings)

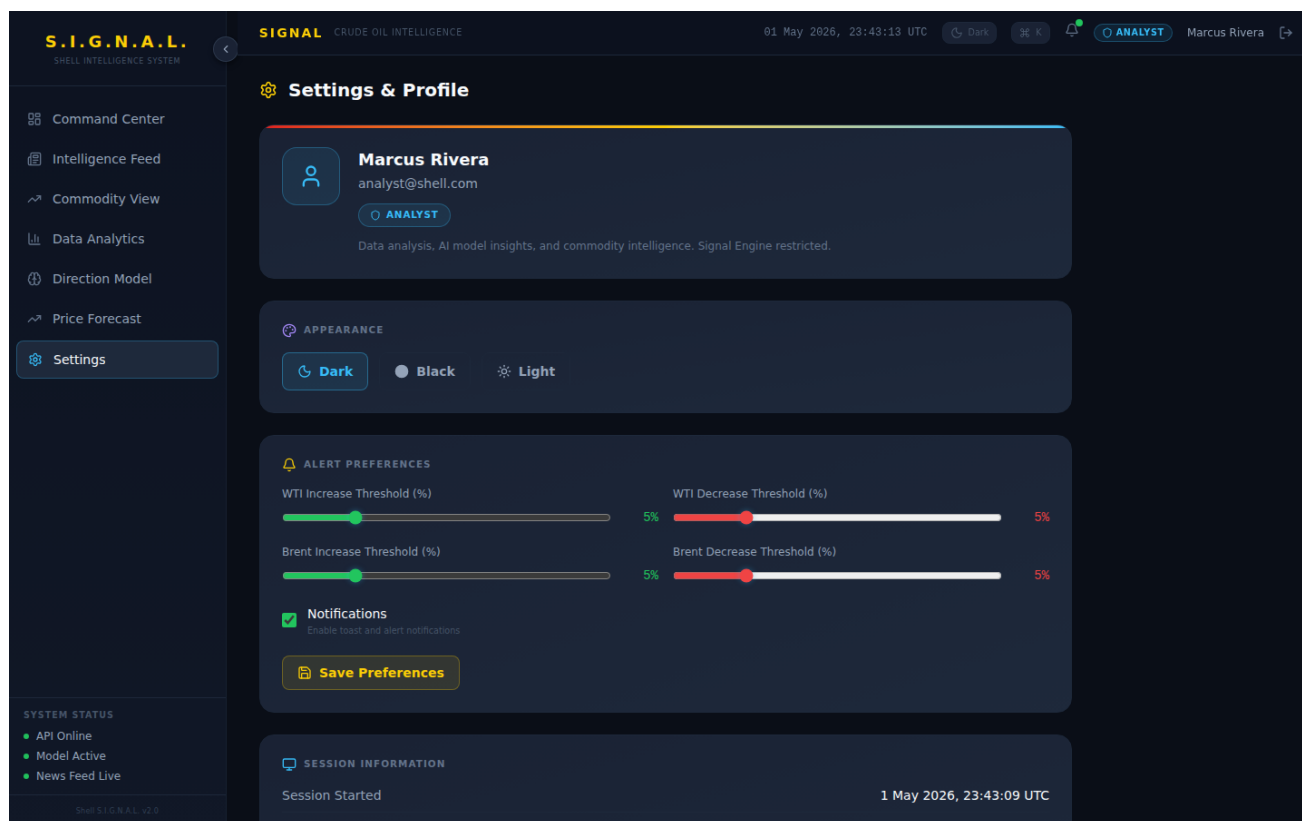


Figure 12-0 — Settings Page Overview

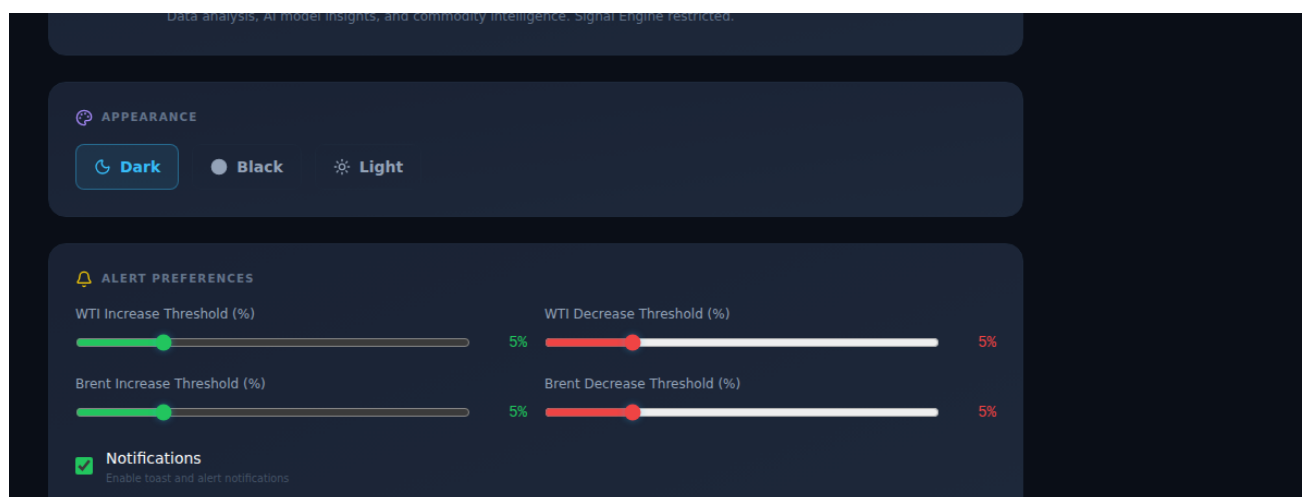


Figure 12-1 — Alert Configuration Panel

Setting Group	Options	Purpose
Appearance	Dark / Black / Light — toggle in TopBar or header	Adapt contrast to ambient lighting and preference
Alert Configuration	Per-commodity price thresholds, sentiment alert controls	Control what triggers a notification; reduce alert fatigue
Executive Config	(Exec only) Briefing cadence and escalation level	Set briefing frequency and severity trigger level
Session	View profile (name, email, role) — Sign Out button	Review identity and securely end session

About	Platform version, model version, last training date	Verify you are running the current model version
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12.2 Themes

Click Moon / Circle / Sun in the TopBar to cycle: **Dark** (charcoal) → **Black** (OLED-optimised) → **Light** (high-contrast). Preference persists in browser storage across sessions.

12.3 Command Palette (■K / Ctrl+K)

Press ■K (Mac) or **Ctrl+K** (Windows) anywhere on the dashboard to open the global Command Palette: navigate to any page, search headlines in real time, or run actions (Refresh Data, Logout) using arrow keys + Enter.

13 Conclusion

SIGNAL represents a mature integration of natural language processing, ensemble machine learning, and financial market data within a purpose-built enterprise interface. It is designed as a high-fidelity decision-support layer — not a replacement for human expertise, but a systematic amplifier of it.

When used correctly — with attention to confidence levels, corroborating news signals, and appropriate risk controls — SIGNAL enables Shell Energy analysts and executives to enter each trading day with a data-driven directional hypothesis backed by 81.5% historical accuracy and a price forecast calibrated to within \$2.07/bbl.

Capability	Performance
Direction accuracy	81.5% on out-of-sample holdout test
Price forecast RMSE	\$2.07/bbl (beats \$2.09 naïve baseline)
Price forecast R ²	0.9532 — 95.3% variance explained
MAPE	~2.5% average daily error
Data latency	Model re-runs daily; dashboard reflects prior close

Support & Feedback:

For model re-training requests, alert configuration help, or data quality issues, contact the SIGNAL development team via the internal Shell Energy IT service portal.

14 Troubleshooting

Symptom	Likely Cause	Resolution
'Backend offline' banner	FastAPI server not running	<code>python -m uvicorn backend.main:app --port 8000</code>
Charts / tables empty	Database empty	Run: <code>python ingest_news.py</code> && <code>python load_price_data.py</code>
Signal shows 'Uncertain' every day	Insufficient recent headlines	Check news ingest schedule; verify NewsAPI key is active
Price forecast shows placeholder values	<code>model_report.json</code> missing	Run: <code>python train_price_model.py</code>
Onboarding skipped for new account	Stale localStorage	Clear browser localStorage or use Incognito/Private mode
Dashboard blank after login	Auth state corrupted	Re-register; select role and access code explicitly

Quick-Start Command Reference

<code>\$ pip install -r requirements.txt</code>	Install all Python dependencies
<code>\$ python -m uvicorn backend.main:app --host 0.0.0.0 --port 8000</code>	Start FastAPI backend
<code>\$ cd frontend && npm install && npm run dev</code>	Start Next.js frontend (dev)
<code>\$ cd frontend && npm run build && npm start</code>	Start Next.js frontend (production)
<code>\$ python train_model.py</code>	Retrain direction classification model
<code>\$ python train_price_model.py</code>	Retrain price regression model
<code>\$ python ingest_news.py</code>	Run news ingestion pipeline